

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination-2023 Course: B. Tech. Branch : Computer Engineering/CSE Semester : VII Subject Code & Name: Big Data Analytics [BTCOE702A] Max Marks: 60 Date:04-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	Give the enabling technologies of Big Data Explain with an Example.	1	6
B)	Draw and explain the Big Data Stack.	1	6
C)	Describe the characteristics of Big Data with an Example.	2	6
Q.2	Solve Any Two of the following.		12
A)	What is HDFS? What are the Features of HDFS?	1	6
B)	Explain the CAP theorem in detail.	1	6
C)	What is Cassandra? Differentiate NoSQL and relational database	2	6
Q.3	Solve Any Two of the following.		12
A)	What is mahout? Explain the features and Applications of Mahout.	2	6
B)	Explain K-Means clustering and Naïve Bayes Classification Algorithm.	2	6
C)	What is Apache Spark? Draw and explain the Components of Spark.	2	6
Q.4	Solve Any Two of the following.		12
A)	What is a Messaging System? Explain the Role and working of KAFKA.	3	6
B)	What is Data Streaming? Explain Big Data Streaming Architecture	3	6
C)	Draw and explain Kappa architecture of real-time Big Data Pipeline	2	6
Q.5	Solve Any Two of the following.		12
A)	What is MongoDB? Write difference between MongoDB and RDBMS.	2	6
B)	What is NoSQL databases? What are the types of NoSQL databases?	2	6
C)	Where we can use MongoDB? Write the Features of MongoDB.	2	6
*** End ***			

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